

Sales Forecasting Analysis & Validation

Historical Revenue 2019–2022 | 2023 FORECAST,ETS Projection | 2022 Backtest Validation

Historical & Forecast Data			
Year	Month	Real Date	Actual Revenue Forecast 2023 Revenue
2019-01	Jan-2019	Jan-2019	14236.895
2019-02	Feb-2019	Feb-2019	4519.892
2019-03	Mar-2019	Mar-2019	55691.009
2019-04	Apr-2019	Apr-2019	28013.973
2019-05	May-2019	May-2019	23648.287
2019-06	Jun-2019	Jun-2019	34595.1276
2019-07	Jul-2019	Jul-2019	33946.393
2019-08	Aug-2019	Aug-2019	27909.4685
2019-09	Sep-2019	Sep-2019	81777.3508
2019-10	Oct-2019	Oct-2019	31453.393
2019-11	Nov-2019	Nov-2019	78628.7167
2019-12	Dec-2019	Dec-2019	69545.6205
2020-01	Jan-2020	Jan-2020	18174.0756
2020-02	Feb-2020	Feb-2020	11951.411
2020-03	Mar-2020	Mar-2020	38726.252
2020-04	Apr-2020	Apr-2020	34195.2085
2020-05	May-2020	May-2020	30131.6865
2020-06	Jun-2020	Jun-2020	24797.292
2020-07	Jul-2020	Jul-2020	28765.325
2020-08	Aug-2020	Aug-2020	36898.3322
2020-09	Sep-2020	Sep-2020	64595.918
2020-10	Oct-2020	Oct-2020	31404.9235
2020-11	Nov-2020	Nov-2020	75972.5635
2020-12	Dec-2020	Dec-2020	74919.5212

Forecast Validation — 2022 Holdout Backtest						
Month	Actual 2022 Revenue	Forecast 2022 Revenue	Forecast Error	Absolute Error		Absolute % Error
Jan-2022	\$43,971	\$63,362	-\$19,391	\$19,391		44.10%
Feb-2022	\$20,301	\$64,419	-\$44,118	\$44,118		217.32%
Mar-2022	\$58,872	\$65,477	-\$6,605	\$6,605		11.22%
Apr-2022	\$36,522	\$66,534	-\$30,013	\$30,013		82.18%
May-2022	\$44,261	\$67,592	-\$23,331	\$23,331		52.71%
Jun-2022	\$52,982	\$68,649	-\$15,668	\$15,668		29.57%
Jul-2022	\$45,264	\$69,707	-\$24,442	\$24,442		54.00%
Aug-2022	\$63,121	\$70,764	-\$7,643	\$7,643		12.11%
Sep-2022	\$87,867	\$71,822	\$16,045	\$16,045		18.26%
Oct-2022	\$77,777	\$72,879	\$4,898	\$4,898		6.30%
Nov-2022	\$118,448	\$73,936	\$44,511	\$44,511		37.58%
Dec-2022	\$83,829	\$74,994	\$8,835	\$8,835		10.54%
MAPE						47.99%
2023 Forecast Summary						
2023 Forecast Revenue	\$860,252					
Average Monthly Forecast	\$71,688					
Highest Forecast Month	Nov-2023		\$109,854			
Lowest Forecast Month	Feb-2023		\$42,569			
Forecast MAPE (2022 Backtest)	47.99%					

Executive Summary						
Overall Performance	Customer Segments	Product Performance	Regional & Shipping Performance	2023 Forecast	Forecast Validation	Recommendation

The business generated approximately \$2.30M in revenue and \$286K in profit from 2019–2022, representing a 12.47% overall profit margin across 5,009 orders.

Consumer was the largest customer segment, generating approximately \$1.16M in revenue and \$134K in profit, followed by Corporate and Home Office.

Technology generated the highest category revenue at approximately \$836K and profit at \$145K. Furniture generated approximately \$742K in revenue but only \$18K in profit, indicating significantly weaker profitability despite strong sales.

The West generated the highest regional revenue at approximately \$725K, followed by the East at \$679K. Average shipping times remained relatively consistent across regions, with Central recording the longest average at approximately 4.06 days.

The ETS model projects approximately \$860K in total 2023 revenue, averaging about \$71.7K per month. November has the highest projected revenue at approximately \$109.9K, while February has the lowest at approximately \$42.6K.

A 2022 holdout backtest produced a MAPE of 47.99%, indicating substantial monthly forecast uncertainty. The 2023 forecast should therefore be interpreted as a directional planning estimate rather than a precise prediction.

Prioritize Technology and the West region as major revenue and profit contributors while investigating Furniture's comparatively low profitability. Additional historical data and model refinement should be considered before relying on the forecast for high-stakes planning decisions.

2021-01	Jan-2021	18542.491	
2021-02	Feb-2021	22978.815	
2021-03	Mar-2021	51715.875	
2021-04	Apr-2021	38750.039	
2021-05	May-2021	56987.728	
2021-06	Jun-2021	40344.534	
2021-07	Jul-2021	39261.963	
2021-08	Aug-2021	31115.3743	
2021-09	Sep-2021	73410.0249	
2021-10	Oct-2021	59687.745	
2021-11	Nov-2021	79411.9658	
2021-12	Dec-2021	96999.043	
2022-01	Jan-2022	43971.374	
2022-02	Feb-2022	20301.1334	
2022-03	Mar-2022	58872.3528	
2022-04	Apr-2022	36521.5361	
2022-05	May-2022	44261.1102	
2022-06	Jun-2022	52981.7257	
2022-07	Jul-2022	45264.416	
2022-08	Aug-2022	63120.888	
2022-09	Sep-2022	87866.652	
2022-10	Oct-2022	77776.9232	
2022-11	Nov-2022	118447.825	
2022-12	Dec-2022	83829.3188	\$83,829
	Jan-2023		\$43,482
	Feb-2023		\$42,569
	Mar-2023		\$70,680
	Apr-2023		\$60,972
	May-2023		\$67,780
	Jun-2023		\$61,895
	Jul-2023		\$63,542
	Aug-2023		\$64,668
	Sep-2023		\$105,726
	Oct-2023		\$64,386
	Nov-2023		\$109,854
	Dec-2023		\$104,698

